

# Multimodal Nowcasting of Sparse Charge-Exchange Spectroscopy on KSTAR: Fast Diagnostics Recover Ion Temperature Beyond Future-Using Interpolation, but Not Toroidal Rotation

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## Abstract

Charge-exchange spectroscopy (CES) is the primary measurement of ion temperature ( $T_i$ ) and toroidal rotation ( $V_{\text{rot}}$ ) in tokamaks, but photon-collection constraints make it slow and intermittently missing: on the KSTAR 10 ms grid studied here,  $\approx 8\%$  of  $T_i$  and  $\approx 24\%$  of  $V_{\text{rot}}$  samples are absent, and the two targets go missing independently. We ask whether the always-available fast diagnostics—beam-emission spectroscopy (BES), electron-cyclotron-emission imaging (ECEI), and Mirnov coils—together with the irregular past CES history can recover information at missing CES timesteps that *temporal interpolation of CES alone cannot*. We train a compact ( $<1\text{M}$ -parameter) multimodal network with per-diagnostic encoders, explicit irregular-time features, per-target masked losses, and physics-informed per-target input routing, and evaluate it against a deliberately adverse bar: offline interpolation (linear, monotone-cubic PCHIP, local AR) that is allowed to see *past and future* CES neighbors, while the model is strictly causal. Under a pre-registered protocol—file-level train/val/test splits, a test set never touched by model selection, and a shot-clustered paired bootstrap (10,000 resamples)—the model beats PCHIP on  $T_i$  with skill  $+0.20$  to  $+0.30$  and a 95% CI excluding zero on *four independent held-out splits*, while  $V_{\text{rot}}$  is statistically indistinguishable from interpolation on all four. Input ablations show this asymmetry is physical rather than architectural: fast diagnostics alone beat persistence for  $T_i$  (collisional  $e-i$  coupling makes  $T_e$  and  $n_e$  informative) but are far worse than the mean for  $V_{\text{rot}}$  (the dominant NBI torque is unobserved and 100 Hz-sampled Mirnov signals alias away kHz mode rotation). The model’s advantage concentrates in high-local-variability neighborhoods where smooth interpolation is weakest. We also document a data-quality audit (54% of observed  $V_{\text{rot}}$  values are forward-filled instrument holds) and an LLM-driven keep/discard architecture-search loop that turned an initially non-significant result into the significant headline. We release the full pipeline, pre-registration, and evaluation artifacts.

## 1 Introduction

Ion temperature  $T_i$  and toroidal rotation  $V_{\text{rot}}$  at the plasma edge govern key elements of tokamak performance and stability, including pedestal structure, ELM behavior, and access to high-confinement regimes. The workhorse measurement of both quantities is charge-exchange spectroscopy (CES) [1], which infers  $T_i$  from Doppler broadening and  $V_{\text{rot}}$  from the Doppler shift of charge-exchange line emission. Because CES must integrate photons to reach adequate signal-to-noise, it is intrinsically slow, and individual time points are frequently lost to exposure and signal-quality problems. In the KSTAR data studied here, even on a uniform 10 ms grid,  $\approx 8\%$  of  $T_i$  and  $\approx 24\%$  of  $V_{\text{rot}}$  observations are missing—*independently* of each other—and, empirically in our dataset, the missingness concentrates at physically interesting moments (low signal-to-noise phases, ELMs, transitions), i.e. it is missing-not-at-random in the sense of Little and Rubin [2] (MNAR).

Meanwhile, several fast diagnostics observe the same plasma without gaps: BES measures density-fluctuation structure, ECEI images electron temperature, and Mirnov coils record magnetic fluctuations. None of them measures  $T_i$  or  $V_{\text{rot}}$  directly. The question this paper asks is:

*At a 10 ms timestep where CES is missing, can the simultaneous fast diagnostics plus the irregular past CES history recover  $T_i$  and  $V_{\text{rot}}$  information that temporal interpolation of CES alone cannot?*

A positive answer would justify a data-driven *virtual sensor*: a model that fills CES gaps online, without the strong inverse-mapping assumptions required by hardware alternatives, enabling gap-free pedestal analysis and, ultimately, real-time use.

**A deliberately hard bar.** The obvious baseline for gap-filling is interpolation of the CES channel itself. We make the comparison intentionally adverse to the model: the interpolation baselines (linear, monotone-cubic PCHIP [3], and a local AR reference) are *offline*—they may use CES neighbors on *both sides* of the target time, including the future—whereas the model is strictly causal, seeing only the fast diagnostics at the target step and a short window of past CES. If a causal model beats a future-using interpolant, the fast diagnostics must carry CES-relevant information that the temporal structure of CES alone does not contain; and a model that beats future-using interpolation a fortiori beats every causal baseline (persistence, AR).

## Contributions.

1. **A leakage-hardened multimodal nowcasting pipeline** for sparse, irregularly observed CES targets: per-target masked multitask losses that recover the  $\approx 28\%$  of labeled rows discarded by both-targets-required filtering; explicit irregular-time features; full masking of the target timestep in the CES-history input; file-level (shot-level) splits; and train-file-only normalization statistics (§3).
2. **A pre-registered, shot-clustered evaluation protocol** (§5) with a three-way split whose test set is never read by model selection, and significance assessed by a shot-clustered paired bootstrap—adjacent samples within a discharge are strongly correlated, so the discharge, not the sample, is the unit of replication.
3. **A robust positive result for  $T_i$**  (§6): the causal model beats future-using PCHIP interpolation with  $\text{skill}_{\text{PCHIP}} = +0.20$  to  $+0.30$  and shot-clustered 95% CIs excluding zero on four independent held-out splits, and survives a genuine-measurement-only re-evaluation.
4. **A physics-grounded negative result for  $V_{\text{rot}}$** , reported as a finding rather than a failure: fast diagnostics carry essentially no rotation information at 10 ms (unobserved NBI torque; Mirnov aliasing), confirmed by input ablations, so  $V_{\text{rot}}$  remains history-autocorrelation-limited and statistically ties interpolation. We call this the  $T_i \leftrightarrow V_{\text{rot}}$  *information asymmetry*.
5. **Diagnosis of where the model earns its edge**: skill is concentrated in high-local-variability neighborhoods, where smooth interpolation is weakest; there, even  $V_{\text{rot}}$  shows a (fragile) significant edge, and an ablation shows the  $T_i$  edge genuinely uses the multimodal signal.
6. **Two methodological artifacts**: a data-quality audit revealing that 54% of observed  $V_{\text{rot}}$  values are forward-filled instrument holds (and a corrected evaluation), and an LLM-driven keep/discard architecture-search loop whose selection gate is the clean interpolation-relative skill rather than validation loss.

## 2 Related work

**Profile fitting and gap-filling in fusion diagnostics.** Gaussian-process regression [4] is the community-standard tool for fitting and uncertainty-quantifying fusion profile data [5, 6], and monotone interpolants such as PCHIP [3, 7] are preferred over natural cubic splines [8] near transients because splines exhibit Gibbs-like ringing at ELM- and sawtooth-like steps [9]. Irregularly sampled series admit exact AR-type

null models (IAR/CIAR) [10, 11]. We use this ladder—persistence, linear interpolation [12], PCHIP, local AR—as pre-registered baselines, with PCHIP as the headline comparator, and Murphy’s MSE skill score [13] as the metric. Our contribution is orthogonal to this literature: rather than a better CES-only interpolant, we test whether *other diagnostics* add information beyond any CES-only method, by benchmarking a causal multimodal model against future-using interpolation.

**Neural diagnostic inference in fusion.** Neural diagnostic inference has three main lines, and it is instructive to ask of each: *what happens when the target measurement is absent?* The oldest line accelerates spectral fitting—NN analysis of JET CX spectra [14, 15] and fast  $T_i$  estimation from EAST CXRS spectra [16]. These gain orders of magnitude in speed over least-squares fits, but by construction they require measured spectra at the timestep: a missing CES exposure leaves them with no input at all. A second line replaces one inference pipeline with another diagnostic’s: Bayesian-model emulation at W7-X [17], profile shapes from scalar parameters on NSTX-U [18], and—closest to our targets— $T_i$  and rotation profiles from an x-ray crystal spectrometer on EAST [19]. Lin et al. predict exactly our two quantities, but from another Doppler spectrometer, i.e. the same physical information channel with the same photon-statistics limits; the mapping is instantaneous and inherits the source diagnostic’s availability. A third line builds end-to-end surrogates for slow reconstruction codes: bolometer tomography [20], line-integral diagnostics [21], safety-factor profiles [22], real-time equilibria [23], and—on KSTAR itself—NN Grad-Shafranov solvers [24, 25] and rapid EPED-parameterized profile reconstruction [26]. These address analysis *latency*, not measurement *availability*: they accelerate inversions of data that exist. Two works sit closer to the availability question. Actuator-conditioned profile forecasting on DIII-D [27] is causal and history-driven like our model, but forecasts from *dense* measured profiles for control, rather than recovering a sparse channel. RTCAKINN [28] reconstructs kinetic profiles (including impurity temperature and rotation) in real time and keeps operating when Thomson/CER inputs drop out, via dropout training; missingness there is a robustness condition handled by zeroing inputs, whereas in our setting the target’s own sparsity is the problem itself, tracked explicitly through per-target observation flags and irregular-time features. Across all three lines the common assumption is that the information source is available when the model runs; recovering a diagnostic at timesteps where *its own* measurement is missing, from other channels plus its past, falls outside their scope.

**Cross-diagnostic reconstruction and temporal densification (closest prior work).** A recent family reconstructs one diagnostic from others. GP-based imputation restores faulty magnetic sensors on KSTAR in sub-ms [29]; ML detects and compensates corrupted sensors [30]; autoencoders fuse redundant density diagnostics for resilience [31]; time-series extrinsic regression fills missing electron-temperature data on EAST [32]; and FusionMAE performs masked-channel “virtual backup diagnosis” across 88 channels on HL-3 [33]. Closest to our framing, Diag2Diag synthesizes Thomson-scattering  $T_e/n_e$  at fluctuation-diagnostic rates from other simultaneous diagnostics on DIII-D [34], and temporal super-resolution of Thomson profiles from fast radiation diagnostics has been demonstrated on COMPASS [35]; PanoMHD models multimodal signal dynamics with a causal transformer for control [36]. Our work differs from this family on four axes, each verifiable from the cited works’ stated scope: (i) *target channel*—every member reconstructs electron quantities or generic dense channels (Thomson  $T_e/n_e$  in 34 and 35; ECE-type  $T_e$  in 32; arbitrary masked channels in 33), and where CER/CES appears at all it is an *input* [34], never the sparse target; (ii) *causality*—Diag2Diag and the COMPASS study are simultaneous, memory-less mappings, and FusionMAE reconstructs within a window, whereas our model conditions on the target’s own irregular past and never reads the future; (iii) *evaluation bar*—none benchmarks against *future-using* interpolation of the target channel, which for gap-filling is the incumbent method to beat, under a pre-registered shot-clustered protocol; and (iv) *information analysis*—reconstructability is assumed rather than tested per target, whereas the  $T_i \leftrightarrow V_{\text{rot}}$  asymmetry is our central finding. Within the surveyed literature we found no causal temporal gap-filling of sparse CXRS ion measurements on any device; we state this as the outcome of our literature search, not as a claim that no such attempt exists.

**Irregular-time and masked multitask learning.** Our design borrows standard ingredients from the irregular time-series literature—observation masks and decay-aware recurrence (GRU-D [37]), attention over continuous-time embeddings (mTAN [38]), and latent-ODE dynamics [39]—together with multitask learning under partial labels [40], late-fusion multimodal architectures [41, 42], pre-LayerNorm transformers [43], and

CES is sparse (targets go missing independently); over half of observed  $V_{\text{rot}}$  values are instrument holds

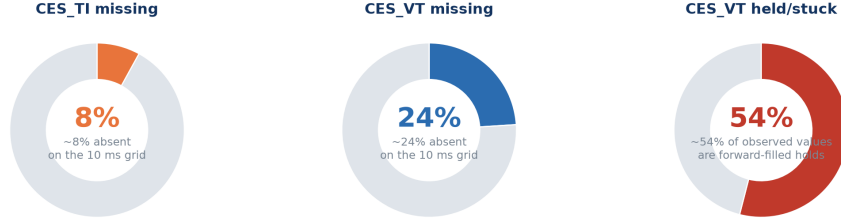


Figure 1: CES missingness structure on the 10 ms grid.  $T_i$  and  $V_{\text{rot}}$  go missing independently ( $\approx 8\%$  /  $\approx 24\%$ ), motivating per-target masking rather than row filtering.

attention pooling [44]. The “nowcasting” framing follows its use in machine-learning weather prediction [45, 46]. The novelty is not any single ingredient but the leakage-hardened combination under a physics-adverse evaluation protocol.

### 3 Data and problem setup

#### 3.1 Dataset

We use 641 KSTAR discharge (shot) files, prioritized from campaigns #24000–#33000 for hardware consistency, targeting H-mode ELM-suppression (RMP) phases; segments of  $\sim 100$  ms are cut around  $D_\alpha$ -active intervals. All signals are aligned to a common 10 ms grid. Each row provides BES [47] (9 channels), ECEI [48] (4 channels), Mirnov coils (2 channels), time, and the two CES targets  $T_i$  and  $V_{\text{rot}}$  [1]. On this grid  $T_i$  is missing in  $\approx 8\%$  of rows and  $V_{\text{rot}}$  in  $\approx 24\%$ , independently (Fig. 1).

#### 3.2 Sample construction

A training sample at target index  $t$  with history window  $W=4$  consists of `bes` ( $W, 9$ ), `ecei` ( $W, 4$ ), `mc` ( $W, 2$ ), `time_features` ( $W, 4$ ), `ces_history` ( $W, 4$ ), the normalized target  $[T_i, V_{\text{rot}}] \in \mathbb{R}^2$ , and a per-target observation mask  $m \in \{0, 1\}^2$ . The four time features—lookback seconds, inter-row delta seconds, and their  $\log(1+x)$  transforms—expose the irregular observation pattern to the model explicitly: the reliability of a past CES value depends strongly on whether it is 10 ms or 200 ms old. The four CES-history channels are the previous normalized  $T_i$ , previous normalized  $V_{\text{rot}}$ , and one observation flag per target (the targets go missing independently, so observation must be tracked per target).

#### 3.3 Design principles

**No fabricated labels.** We never impute targets to create training rows. A row is kept iff its diagnostic inputs are complete and *at least one* CES target is genuinely observed.

**Per-target masked loss.** The loss is masked per target,  $\mathcal{L} = \sum_{k \in \{T_i, V_{\text{rot}}\}} m_k (\hat{y}_k - y_k)^2 / \sum_k m_k$ , so a row with only one observed target still supervises that target. The previous both-targets-required filter silently discarded  $\approx 28\%$  of labeled rows; removing it is a pure data gain.

**Leakage hardening.** (i) *File-level splits*: adjacent rows within a shot are strongly autocorrelated, so splits are made at the shot-file level and pinned to disk; (ii) *train-file-only normalization*: all  $z$ -score statistics (NaN-aware for the sparse targets) are estimated on training files only; (iii) *target-step masking*: in `ces_history`, the target timestep’s values *and* observation flags are zeroed, so the model cannot read its own answer.

### 3.4 Data-quality audit: held (forward-filled) $V_{\text{rot}}$ values

A late audit found that **54% of observed  $V_{\text{rot}}$  values are instrument holds**: bit-identical repeats of the previous observation within a contiguous time block (runs up to 1,214 rows; 499/641 files affected). These arise because the native  $V_{\text{rot}}$  cadence is slower than the row cadence; they are not independent measurements.  $T_i$  is unaffected (0.0%). A controlled A/B retrain showed training is *not* contaminated—masking holds at train time does not improve (and slightly hurts) genuine-measurement performance, i.e. the holds act as a harmless persistence prior—so holds are kept in training but *excluded from evaluation*. The correction inflates reported physical  $V_{\text{rot}}$  RMSE by 35–55% (from 22–33 to 35–46 in raw units) because held targets have near-zero baseline error; all conclusions below use the corrected (genuine-only) evaluation, and the  $T_i$  headline is unchanged by it.

## 4 Model

The predictor is a compact (<1M parameters) late-fusion network; the bottleneck in this problem is generalization, not capacity.

**Per-diagnostic encoders.** Each modality (BES, ECEI, MC) is encoded by its own time-aware 1-D CNN that consumes the modality jointly with the time features and history, preserving each diagnostic’s channel structure until fusion. CNNs are preferred over recurrence because gap-removal makes the effective sampling irregular, violating the uniform-step assumption RNNs rely on.

**History encoder.** The CES history is encoded by a 2-layer pre-LayerNorm Transformer [43] with learned positional embeddings and *multi-head attention pooling* [44] that outputs both an attention-weighted mean and an attention-weighted standard deviation; the latter exposes “how locally variable is the recent history,” which §6.5 shows is where the model earns its edge. Self-attention over the (masked) target step and its observed neighbors performs an operation analogous to learned interpolation, with per-target observed-neighbor biases.

**Physics-informed per-target routing.** The  $T_i$  head fuses fast diagnostics + time + history; the  $V_{\text{rot}}$  head fuses *history and time only*, deliberately excluding the fast diagnostics, which ablations (§6.4) show carry no usable rotation information at 10 ms. Each target has its own pre-LayerNorm MLP head; outputs remain in normalized units (denormalization happens only in evaluation).

**Search lessons.** Across  $\sim 40$  controlled architecture iterations, pre-LayerNorm was the stability turning point (validation loss  $0.89 \rightarrow 0.49$ ), attention pooling was the most reliable single improvement, and aggressive capacity scaling, complex skip topologies, and extra convolutional extractors consistently failed to improve held-out performance.

## 5 Evaluation methodology

**Metric.** All errors are denormalized to physical CES units and computed per target. The primary score is Murphy’s skill [13] against the pre-registered headline baseline,  $\text{skill}_{\text{PCHIP}} = 1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{PCHIP}}$  (positive = model better; 0 = tie). Every arm (model and all baselines) is scored on the *identical* (file, row) sample set under the same per-target keep mask. Interpolants exclude the target’s own value (no leakage) and refuse to interpolate across gaps  $\geq 0.5$  s, falling back to persistence.

**Three-way split; selection never sees TEST.** Files are split train/val/test; the test set was reserved *before* any architecture search and is never read by the search loop, so headline numbers carry no winner’s-curse bias. The canonical test set contains 34,644 scored samples across 96 shots (observed  $n$ :  $T_i$  32,716,  $V_{\text{rot}}$  27,437).

Table 1: Physical-unit RMSE ladder on the canonical held-out test split (pre-improvement baseline model; the final model improves further, see §6.2). Lower is better.  $V_{\text{rot}}$  RMSE values here predate the hold correction of §3.4; corrected genuine-only  $V_{\text{rot}}$  RMSE is 35–46.

| Arm                      | Information access    | $T_i$ RMSE   | $V_{\text{rot}}$ RMSE |
|--------------------------|-----------------------|--------------|-----------------------|
| <b>Model (nowcaster)</b> | fast diag. + past CES | <b>412.4</b> | <b>23.2</b>           |
| Linear interpolation     | past + future CES     | 422.7        | 24.0                  |
| PCHIP interpolation      | past + future CES     | 431.8        | 24.5                  |
| Persistence              | last observed CES     | 487.3        | 27.8                  |
| AR (local, causal)       | past CES only         | 1005.7       | 57.2                  |

**Pre-registration.** Four rules were fixed in writing before viewing test numbers: **PR1** the headline comparator is PCHIP (chosen a priori for monotonicity/no ringing at transients; the full ladder {persistence, linear, PCHIP, AR} is also reported); **PR2** interpolation predicts everywhere the model is scored, with persistence fallback where no future neighbor exists (no population thinning); **PR3** a minimum test-set floor ( $\geq 15$  shots,  $\geq 3,000$  observed  $T_i$  samples; met); **PR4** significance means a shot-clustered bootstrap 95% CI on the skill excluding zero.

**Shot-clustered paired bootstrap.** Adjacent CES rows within a discharge are strongly correlated; treating samples as independent would grossly understate uncertainty. Per-sample paired errors  $\text{SE}_{\text{model}} - \text{SE}_{\text{PCHIP}}$  are aggregated by shot, and whole shots are resampled with replacement (10,000 resamples, fixed seed). The resulting CI reflects genuine shot-to-shot generalization; its cost is that the effective sample size is the number of shots ( $\approx 96$ ), which bounds statistical power throughout.

**Honest caveat (MNAR).** Truly missing points have no ground truth, so skill is measured by masking *observed* points. Because CES missingness is MNAR (drop-out at low S/N, ELMs, transitions), observed-point skill is an *optimistic bound* for performance at genuinely missing points. We state this rather than hide it.

## 6 Results

### 6.1 The model dominates every causal baseline

Table 1 shows the physical-unit RMSE ladder on the canonical test split. The model has the lowest RMSE for both targets and beats the causal baselines—persistence and past-only AR, i.e. everything actually available in real time—by a wide margin. This is the most robust result in the paper: for any online setting, where future CES is by definition unavailable, the model is the clear winner.

### 6.2 Headline: $T_i$ beats future-using interpolation on four independent splits

With the final architecture (GRU history encoder + per-target multi-head-attention heads,  $W=4$ ; selected by the search loop of §7 on validation skill only), we evaluated four *independent* held-out test splits (seeds 42, 1, 7, 123; the latter three were never used in any selection step):

- $T_i$ : **PASS on all four splits.**  $\text{skill}_{\text{PCHIP}} = +0.270$  (95% CI  $[+0.144, +0.364]$ ),  $+0.200$ ,  $+0.271$ ,  $+0.295$ ; every shot-clustered CI excludes zero, against both PCHIP and linear. The result survives genuine-measurement-only re-evaluation ( $+0.225/ +0.204/ +0.285/ +0.288$ , all PASS).
- $V_{\text{rot}}$ : **not significant on any split.** Point estimates are positive but every CI includes zero. We claim a tie, not a win.

An honest progression note: the initial baseline model was *not* significant (single split,  $+0.088$ , n.s.). Switching the architecture-search selection gate from validation loss to the clean interpolation-relative skill produced the final model, which is significant on all four splits (Fig. 4). We report both stages.



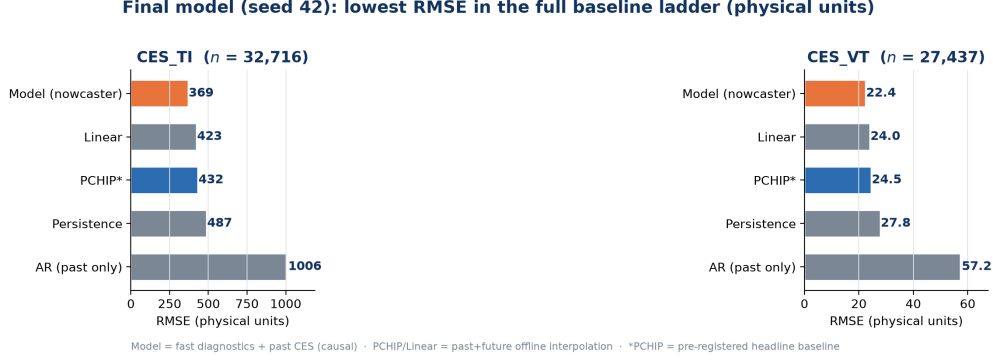


Figure 2: RMSE ladder for the *final* model (seed 42, test split): the causal model attains the lowest RMSE for both targets, decisively beating the causal baselines and edging the future-using interpolants on the point estimate. Table 1 reports the same ladder for the earlier baseline model, hence the different absolute values.

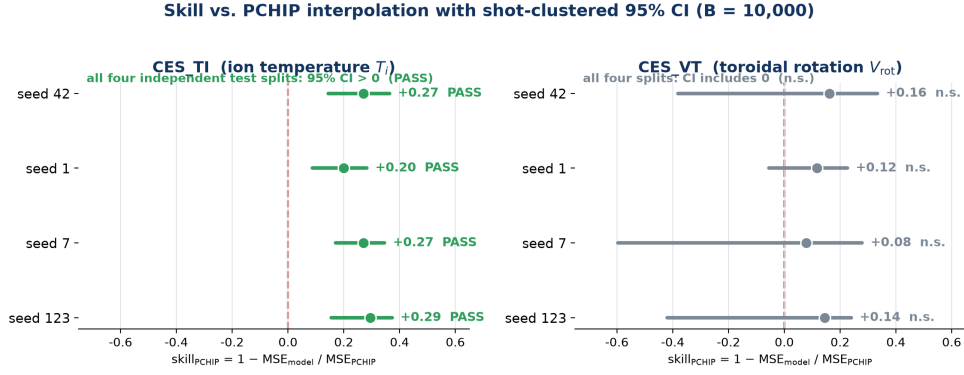


Figure 3: Forest plot of skill<sub>PCHIP</sub> with shot-clustered 95% CIs across four independent held-out splits.  $T_i$  excludes zero on all four;  $V_{rot}$  on none.

### 6.3 Where the skill lives: small gaps dominate

Stratifying by the time since the last observed CES value,  $\geq 97\%$  of real targets lie in the small-gap regime ( $\Delta t \leq 15$  ms), where the model wins ( $T_i + 0.08$ ,  $V_{rot} + 0.23$  vs. PCHIP on the canonical split). At very large gaps ( $> 105$  ms) the future-anchored interpolant dominates, but those bins hold only tens of samples and carry no per-bin CIs; they are not load-bearing. The one well-powered comparison—small gaps, which is precisely where nowcasting is needed—favors the model.

### 6.4 The $T_i \leftrightarrow V_{rot}$ information asymmetry

A controlled input-modality ablation (identical training budget; one modality group zeroed at a time; persistence always computed from real history) explains *why* the two targets behave differently (Table 2, Fig. 5).

- **$T_i$ : fast diagnostics carry real information.** Fast-only still beats persistence (+0.16). The physical channel is collisional electron-ion coupling ( $t_{ei} \propto T_e^{3/2}/n_e$ ): ECEI supplies  $T_e$  and BES supplies  $n_e$  structure.
- **$V_{rot}$ : information is almost entirely CES history.** Fast-only scores  $-3.31$  (worse than predicting the mean). Toroidal rotation is governed by momentum sources that none of our inputs observe—the external NBI torque and intrinsic rotation drives [49]—and the Mirnov signals are sampled at 100 Hz, aliasing away the kHz mode-rotation frequencies that could otherwise proxy rotation.

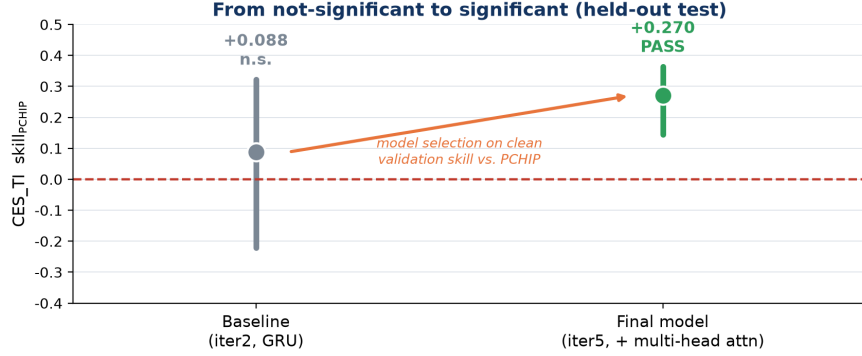


Figure 4: From non-significant (baseline, +0.088 n.s.) to significant (final model, +0.20–+0.30, PASS  $\times 4$ ): the decisive change was selecting architectures on clean skill rather than validation loss.

Table 2: Input-modality ablation (validation split, skill vs. persistence, physical units). Fast diagnostics alone carry genuine  $T_i$  information but are catastrophically uninformative for  $V_{\text{rot}}$ .

| Model inputs                                | $T_i$ skill | $V_{\text{rot}}$ skill |
|---------------------------------------------|-------------|------------------------|
| Full (history + fast + time)                | +0.36       | +0.18                  |
| History only ( <i>no fast</i> )             | +0.39       | +0.23                  |
| Fast diagnostics only ( <i>no history</i> ) | +0.16       | −3.31                  |

This asymmetry was predicted from physics before the ablation and is confirmed by it; the  $V_{\text{rot}}$  non-win is a finding about diagnostic information content, not a model failure. It also motivates the per-target routing of §4.

## 6.5 The edge concentrates in high-variability neighborhoods

Using a target-independent, input-side proxy for local activity (large neighbor-bracket slope and/or high local CES-neighbor variance, computed *excluding* the target row), we split evaluation into “peak” (high-local-activity) and bulk regions (validation split, shot-clustered CIs; Fig. 6):

- $T_i$ : global +0.52  $\rightarrow$  peak **+0.86** (CI [+0.66, +0.93], PASS). Interpolation is near-optimal on the smooth bulk; the model’s value is in the active stretches.
- $V_{\text{rot}}$ : global +0.24  $\rightarrow$  peak **+0.69** (CI [+0.11, +0.87], PASS, robust across three sensitivity settings but with a thin lower bound). The asymmetry is therefore *regional*: globally  $V_{\text{rot}}$  ties interpolation, but in high-activity neighborhoods — where smooth past+future interpolation is worst — even the history-driven  $V_{\text{rot}}$  predictor adds value. We report this as a regional strength on the optimism-caveated validation split, not a reversal of the global null result.
- A peak-restricted ablation (zeroing BES/ECEI/MC) hurts  $T_i$  significantly (paired-difference CI  $\gg 0$ ) but leaves  $V_{\text{rot}}$  unchanged: the  $T_i$  peak edge genuinely uses the multimodal signal, while the  $V_{\text{rot}}$  peak edge is history-driven.

## 7 Search method: an LLM-driven keep/discard loop

The final architecture was found by an automated research loop in which an LLM plays the researcher role (inspired by autoresearch-style keep/discard protocols). Each iteration: (i) smoke-validate and fully train the current proposal; (ii) score it on the *clean, non-augmented* validation skill<sub>PCHIP</sub>—never the augmented validation loss, and never any test split; (iii) **keep** the change iff it improves the best score so far, else **discard**



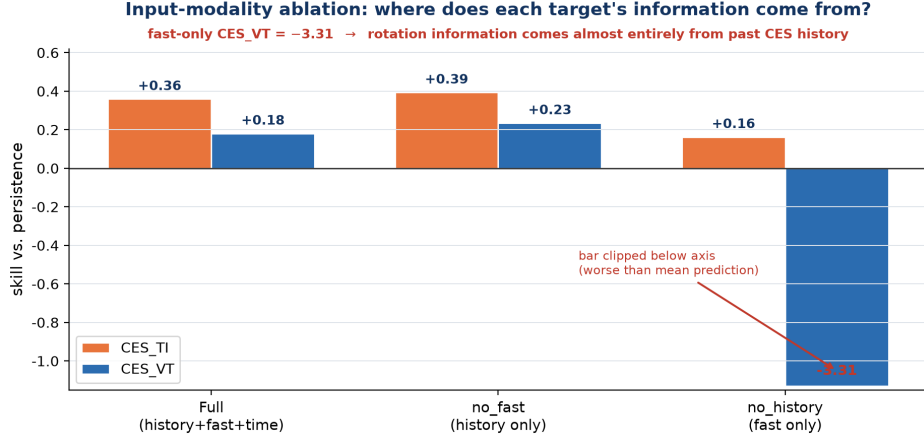


Figure 5: Modality ablation: fast-only beats persistence for  $T_i$  but is far worse than the mean for  $V_{\text{rot}}$  — the information asymmetry.

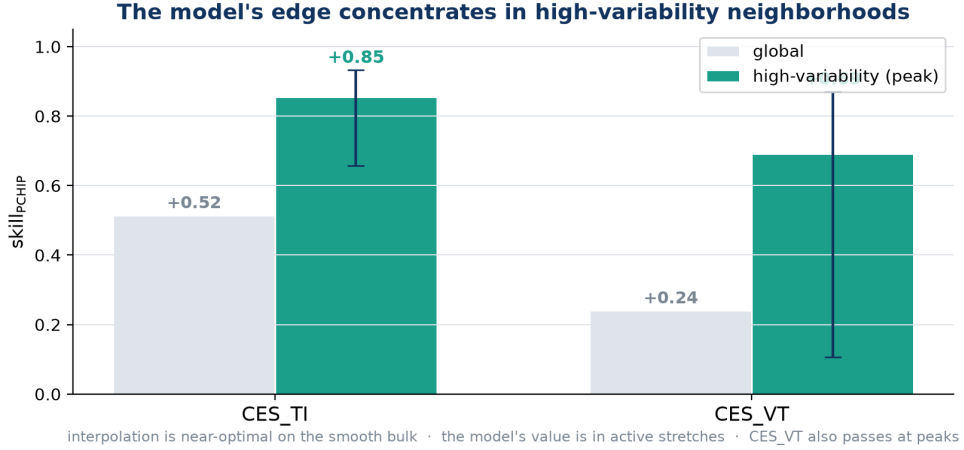


Figure 6: Global vs. high-variability (“peak”) skill with shot-clustered 95% CIs. The model’s advantage over interpolation concentrates where interpolation is weakest;  $V_{\text{rot}}$  passes at peaks despite the global tie.

and roll back `model.py` to the best snapshot; (iv) brief the LLM with the run history, a hard data contract (interfaces, masking, and leakage rules the proposal must preserve), and accumulated project knowledge, and request *one controlled change*. The loop therefore never builds on a regression, and training failures are treated as repair signals rather than evidence. Its concrete contribution here was the n.s.→significant progression of Fig. 4; we document it as reproducibility infrastructure, not as the scientific contribution.

## 8 Limitations

**Statistical power.** The unit of replication is the shot; with  $\approx 96$  ( $T_i$ ) / 91 ( $V_{\text{rot}}$ ) test shots, power bounds every significance call, and per-shot squared-error differences are heavy-tailed (a few discharges dominate the bootstrap spread). **MNAR optimism.** Skill is measured on observed points only; genuinely missing points may be harder. **Metric asymmetry.** Interpolants use full-shot neighbors while the model sees a  $W=4$  history; this is deliberately adverse to the model but complicates direct interpretation. **Single architecture and window.** Sensitivity to  $W$  and architecture is uncharacterized; results are for one selected model. **Large-gap bins** contain tens of samples without CIs and are unreliable. **Scope.** The pedestal-top framing is inherited from the data selection (pedestal-top CES channels); no radial-dependence experiment

is performed, and event-phase analyses (ELM cycle, L–H transitions) are future work—the high-variability analysis of §6.5 is an event-agnostic proxy.

## 9 Conclusion

We built a causal, leakage-hardened multimodal nowcaster for sparse KSTAR CES targets and evaluated it under a pre-registered, shot-clustered protocol against a deliberately adverse bar: interpolation that sees the future. The model decisively beats every causal baseline; for ion temperature it also beats future-using PCHIP interpolation with significant, replicated skill (+0.20–+0.30; four independent splits), while toroidal rotation ties interpolation for a physically identified reason—fast diagnostics simply do not carry rotation information at 10 ms. The advantage concentrates exactly where gap-filling matters: small gaps and high-variability neighborhoods. Beyond the headline, the work contributes a corrected  $V_{\text{rot}}$  evaluation (54% instrument holds), an information-asymmetry ablation, and an LLM-driven keep/discard search protocol. Future work targets the power limitation (more test shots, multi-campaign data), bracket-distance stratification to isolate the regime where fast-diagnostic information has maximal marginal value, peak-weighted training, and rotation-side improvements that better exploit history—since more sensors, demonstrably, will not help  $V_{\text{rot}}$  at this timescale.

### Reproducibility

Splits are pinned to disk and verified on reload; normalization statistics are train-file-only; the search loop never reads test data; bootstrap uses a fixed seed and 10,000 resamples; all headline numbers are read from frozen evaluation artifacts. Code and evaluation artifacts accompany the submission; the public repository link will be inserted at camera-ready.

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